Substitution and Rearranging Formulae

Name _____ **Date** _____ No calculator Version 1

KEY WORDS substitute · expression · formula · subject · rearrange · inverse operation

A Substituting positive numbers ●○○

ANSWER ON THE SHEET

Remember: Two letters side by side are multiplied: ab means $a \times b$, and $3a$ means $3 \times a$.

1. Find the value of each expression when $a = 3$, $b = 5$ and $c = 2$.

a) $2a + b$ = _____ **b)** ab = _____ **c)** $b^2 - a$ = _____ **d)** $3(a + b)$ = _____

e) abc = _____ **f)** $\dfrac{b + 7}{a}$ = _____ **g)** $2a^2$ = _____ **h)** $a^2 + c^2$ = _____

i) $4b - 3c$ = _____ **j)** $\dfrac{ab + 1}{c}$ = _____ **k)** $(a + c)^2$ = _____ **l)** $c^3 - b$ = _____

B Negative numbers and powers ●●○

WORK IN YOUR BOOK

Watch out: Put brackets around a negative number when you substitute it: $(-3)^2 = 9$, not -9.

2. Find the value of each expression when $p = -2$, $q = 4$ and $r = -3$.

a) $p + q$ **b)** pq **c)** $q - r$ **d)** p^2

e) pr **f)** $2p + 3r$ **g)** $r^2 - q$ **h)** $2p^2 + r$

i) pqr **j)** $\dfrac{q - p}{r}$ **k)** $q^2 - p^2$ **l)** $3r^2 - 2q$

C Using formulae ●●○

WORK IN YOUR BOOK

3. Substitute into each formula.

a) $v = u + at$. Find v when $u = 5$, $a = 3$ and $t = 4$.

b) $A = \dfrac{1}{2}bh$. Find A when $b = 8$ and $h = 5$.

c) Celsius to Fahrenheit: $F = 1.8C + 32$. Find F when and $C = 25$.

d) $s = ut + \dfrac{1}{2}at^2$. Find s when $u = 4$, $a = 6$ and $t = 3$.

e) $P = 2(l + w)$. Find P when $l = 7.5$ and $w = 4$.

f) Area of a trapezium: $A = \dfrac{h(a + b)}{2}$. Find A when $h = 6$, $a = 5$ and $b = 9$.

g) $E = \dfrac{1}{2}mv^2$. Find E when $m = 4$ and $v = 5$.

h) A plumber charges C dollars for h hours: $C = 40 + 25h$. Find C when and $h = 3$.

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D Rearranging: one step ●●○ WORK IN YOUR BOOK

Remember: To make a letter the subject, undo what has been done to it. Do the same to both sides.

4. Make the letter in square brackets the subject of the formula.

a) $y = x + 5 \quad [x]$

b) $A = lw \quad [w]$

c) $P = 4s \quad [s]$

d) $v = \dfrac{d}{t} \quad [d]$

e) $y = x - c \quad [x]$

f) $C = 2\pi r \quad [r]$

g) $F = ma \quad [a]$

h) $m = \dfrac{k}{5} \quad [k]$

i) $V = lwh \quad [h]$

E Rearranging: two steps ●●● WORK IN YOUR BOOK

Remember: Undo adding and subtracting first. Then undo multiplying and dividing.

5. Make the letter in square brackets the subject of the formula.

a) $y = mx + c \quad [x]$

b) $v = u + at \quad [t]$

c) $A = \dfrac{1}{2}bh \quad [h]$

d) $P = 2l + 2w \quad [l]$

e) $F = 1.8C + 32 \quad [C]$

f) $y = \dfrac{x}{3} - 2 \quad [x]$

g) $y = 5(x + a) \quad [x]$

h) $T = \dfrac{a + b}{2} \quad [a]$

i) $y = k - 3x \quad [x]$

★ Extension FOR EARLY FINISHERS

6. These formulae have squares and square roots. Make the letter in square brackets the subject.

a) $A = \pi r^2 \quad [r]$

b) $E = \dfrac{1}{2}mv^2 \quad [v]$

c) $v^2 = u^2 + 2as \quad [s]$

d) $y = \sqrt{x + a} \quad [x]$

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Substitution and Rearranging Formulae

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS substitute · expression · formula · subject · rearrange · inverse operation

A Substituting positive numbers ●○○ — ANSWER ON THE SHEET

Remember: Two letters side by side are multiplied: ab means $a \times b$, and $3a$ means $3 \times a$.

1. Find the value of each expression when $a = 4$, $b = 6$ and $c = 3$.

a) $3a + b$ = _____ **b)** ac = _____ **c)** $b^2 - c$ = _____ **d)** $2(a + c)$ = _____

e) abc = _____ **f)** $\dfrac{b + 6}{c}$ = _____ **g)** $3c^2$ = _____ **h)** $a^2 + b^2$ = _____

i) $5a - 2b$ = _____ **j)** $\dfrac{ab + 3}{c}$ = _____ **k)** $(b - c)^2$ = _____ **l)** $c^3 - a$ = _____

B Negative numbers and powers ●●○ — WORK IN YOUR BOOK

Watch out: Put brackets around a negative number when you substitute it: $(-3)^2 = 9$, not -9.

2. Find the value of each expression when $p = -3$, $q = 5$ and $r = -2$.

a) $p + q$ **b)** pr **c)** $q - p$ **d)** r^2

e) pq **f)** $3p + 2r$ **g)** $p^2 - q$ **h)** $2r^2 + p$

i) pqr **j)** $\dfrac{q - p}{r}$ **k)** $q^2 - r^2$ **l)** $2p^2 - 3q$

C Using formulae ●●○ — WORK IN YOUR BOOK

3. Substitute into each formula.

a) $v = u + at$. Find v when $u = 8$, $a = 2$ and $t = 6$.

b) $A = \dfrac{1}{2}bh$. Find A when $b = 10$ and $h = 7$.

c) Celsius to Fahrenheit: $F = 1.8C + 32$. Find F when and $C = 35$.

d) $s = ut + \dfrac{1}{2}at^2$. Find s when $u = 5$, $a = 4$ and $t = 3$.

e) $P = 2(l + w)$. Find P when $l = 6.5$ and $w = 3$.

f) Area of a trapezium: $A = \dfrac{h(a + b)}{2}$. Find A when $h = 4$, $a = 7$ and $b = 11$.

g) $E = \dfrac{1}{2}mv^2$. Find E when $m = 6$ and $v = 3$.

h) An electrician charges C dollars for h hours: $C = 60 + 35h$. Find C when and $h = 4$.

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D Rearranging: one step ●●○ WORK IN YOUR BOOK

Remember: To make a letter the subject, undo what has been done to it. Do the same to both sides.

4. Make the letter in square brackets the subject of the formula.

a) $y = x + 8 \quad [x]$

b) $A = bh \quad [h]$

c) $P = 6s \quad [s]$

d) $s = \dfrac{d}{t} \quad [d]$

e) $y = x - k \quad [x]$

f) $C = \pi d \quad [d]$

g) $V = IR \quad [I]$

h) $n = \dfrac{t}{4} \quad [t]$

i) $V = lwh \quad [l]$

E Rearranging: two steps ●●● WORK IN YOUR BOOK

Remember: Undo adding and subtracting first. Then undo multiplying and dividing.

5. Make the letter in square brackets the subject of the formula.

a) $y = mx - c \quad [x]$

b) $v = u + at \quad [a]$

c) $A = \dfrac{1}{2}bh \quad [b]$

d) $P = 2l + 2w \quad [w]$

e) $C = 40 + 25h \quad [h]$

f) $y = \dfrac{x}{4} + 3 \quad [x]$

g) $y = 2(x - b) \quad [x]$

h) $M = \dfrac{p + q}{2} \quad [q]$

i) $y = n - 5x \quad [x]$

★ Extension FOR EARLY FINISHERS

6. These formulae have squares and square roots. Make the letter in square brackets the subject.

a) $V = \pi r^2 h \quad [r]$

b) $A = 4\pi r^2 \quad [r]$

c) $v^2 = u^2 + 2as \quad [a]$

d) $y = \sqrt{x - b} \quad [x]$

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